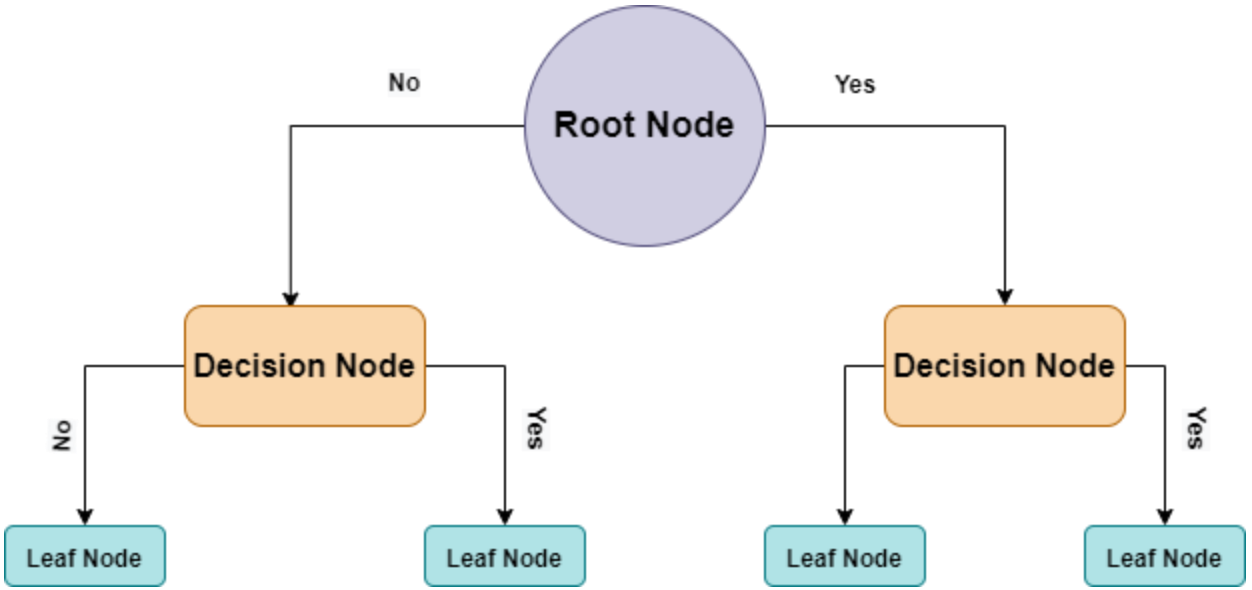
ITEC 4220 Adv Data Analytics (Modified from CIS4930: Data Science Foundations)

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Recap: Decision Tree

Decision Trees

- Decision tree is a simple classifier
  - ✓A flow-chart-like tree structure



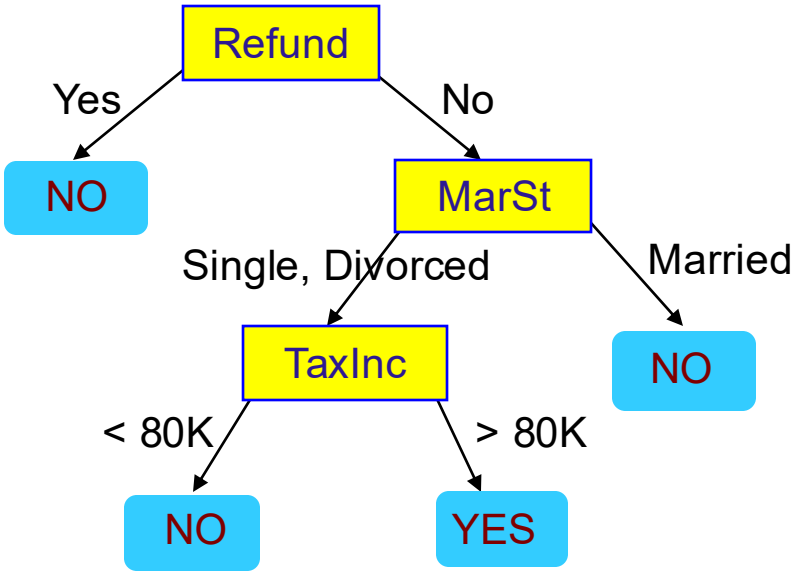
Trees & Rules

- Trees are based on separating records into subgroups by creating splits on predictors
- These splits create logical **rules**
  - ✓**For example:** IF Age<55 AND Education>12 THEN Class=1

Example

Tid	Refund	Marital Status	Taxable Income	Cheat
	categorical	categorical	continuous	class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Training Data



Model: Decision Tree

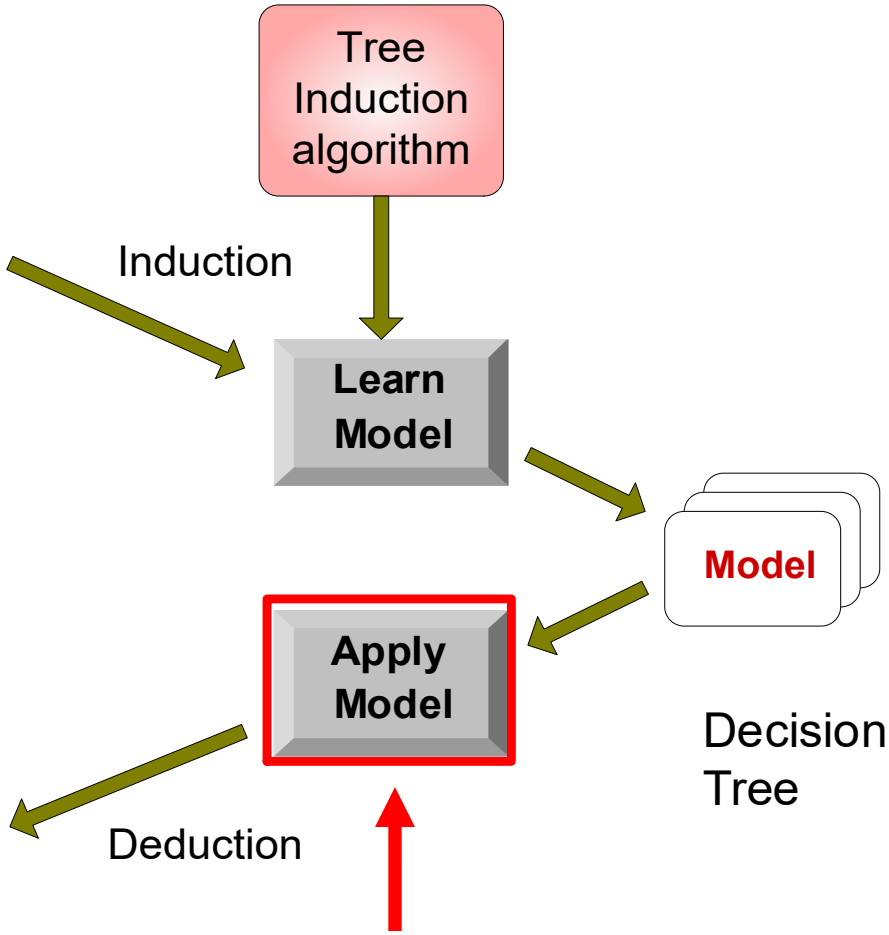
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

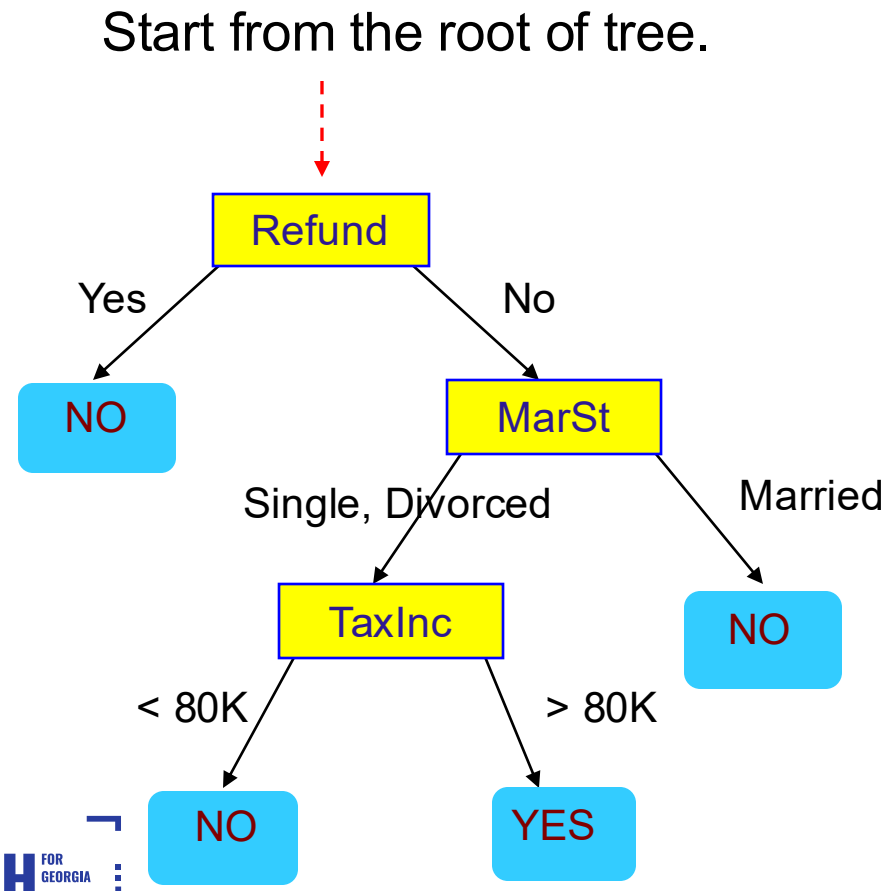
Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Apply Model to Test Data



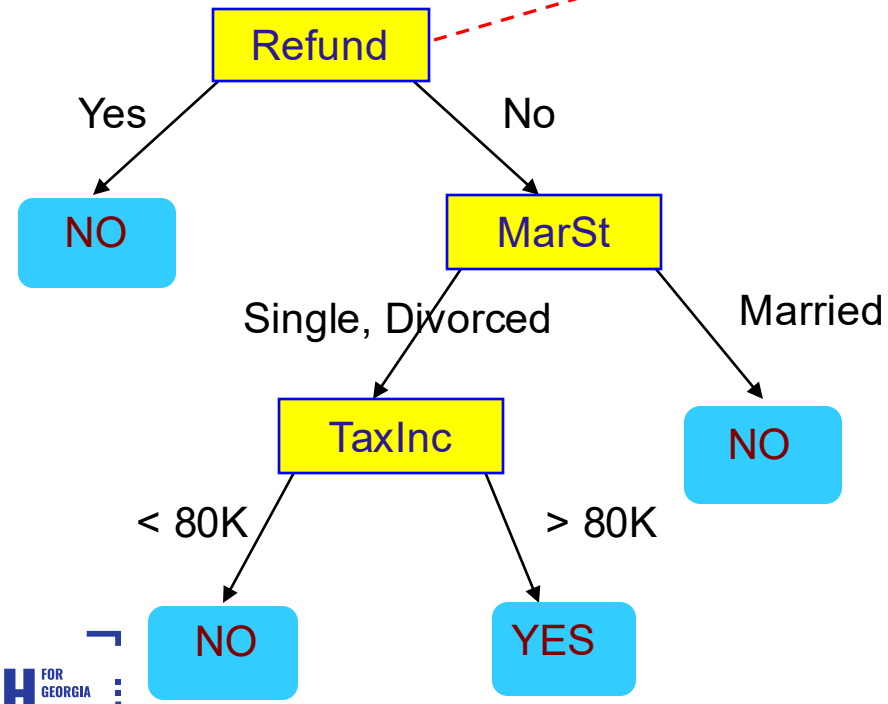
Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

Apply Model to Test Data

Test Data

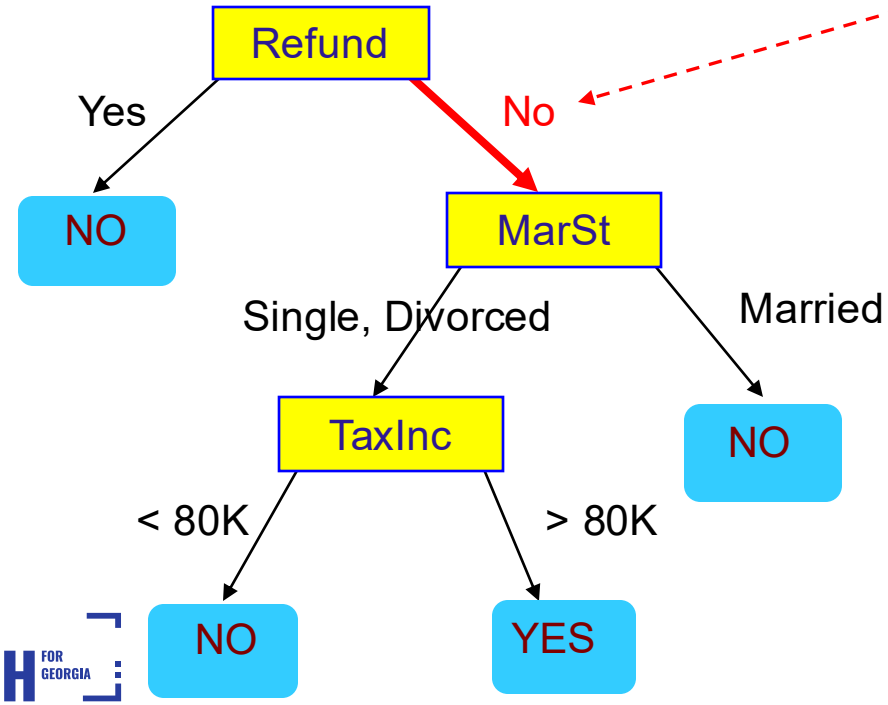
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

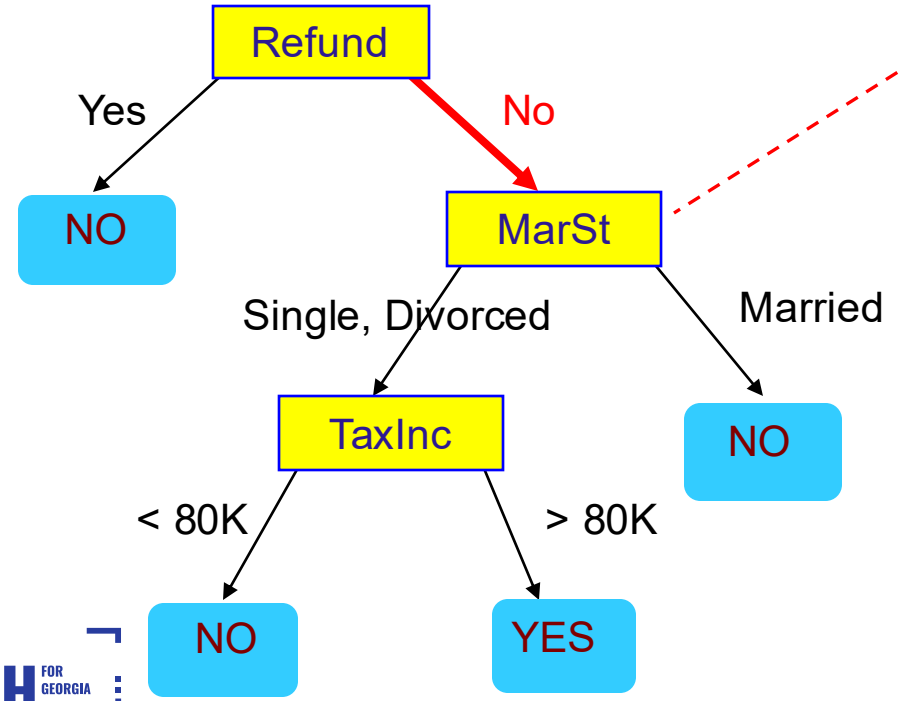
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

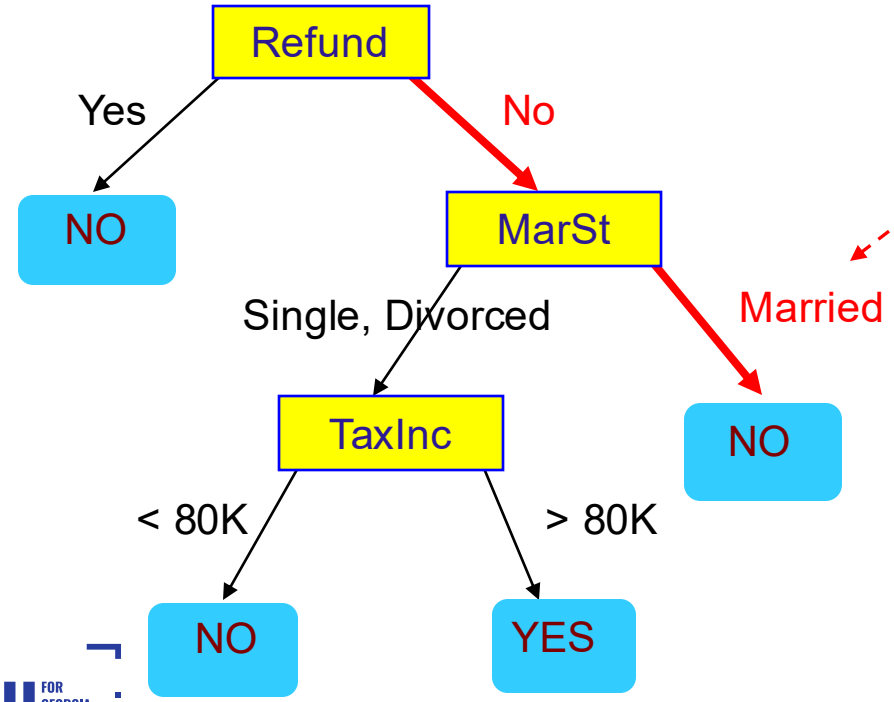
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

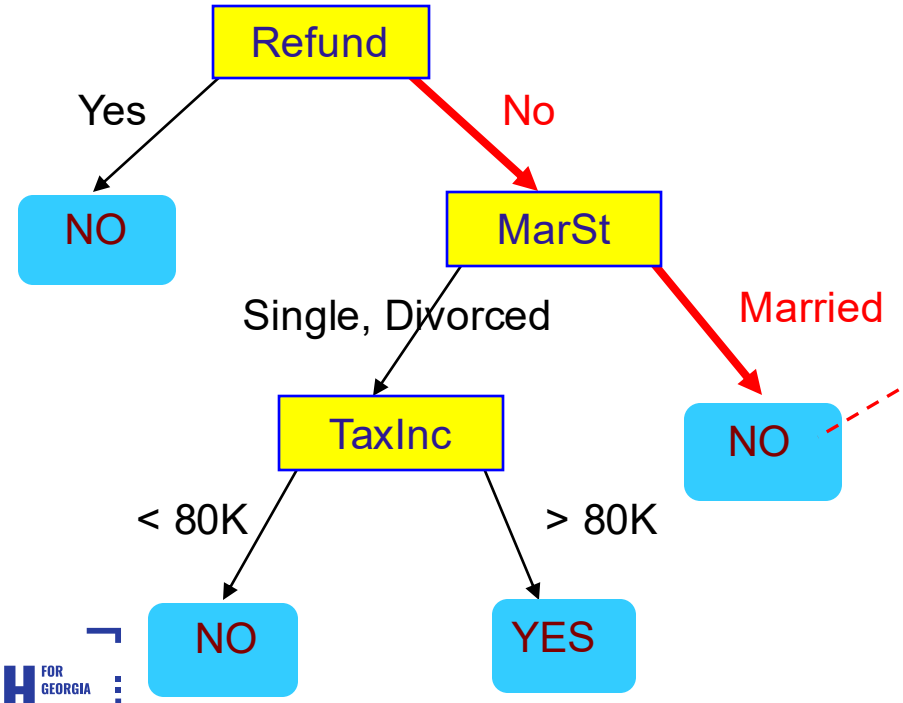
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Assign Cheat to “No”

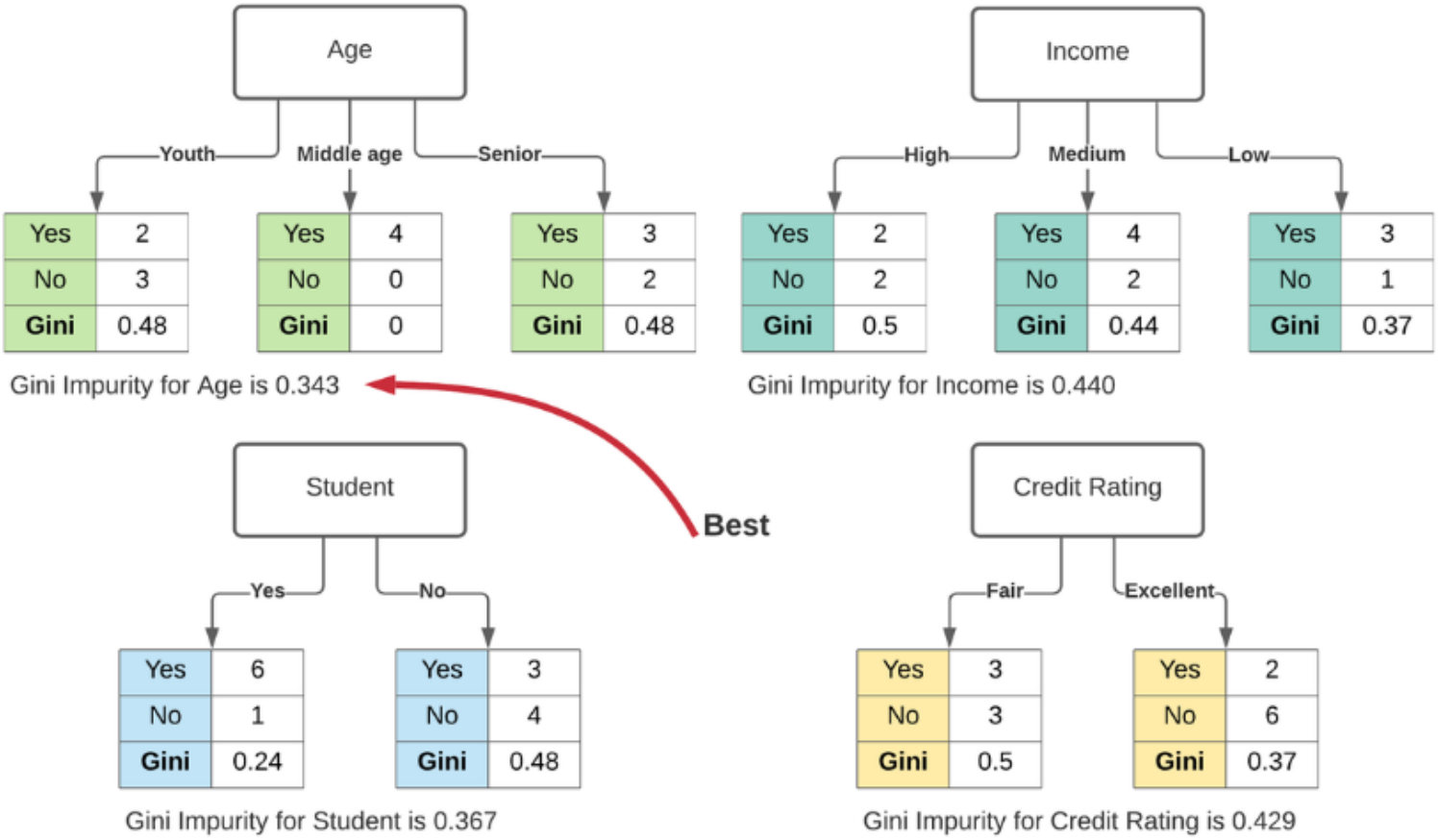
How Is a Tree Produced?

- **Recursive partitioning:** Repeatedly split the records into two parts to achieve maximum homogeneity of outcome within each new part!
- Homogeneity, purity, means how uniform the data in a given node of a decision tree is.
- A node is considered pure if all the data points in it belong to the same class.

Impurity Measurements

- Gini Index
- Entropy Measure

Impurity & Recursive Partitioning



Random Forest

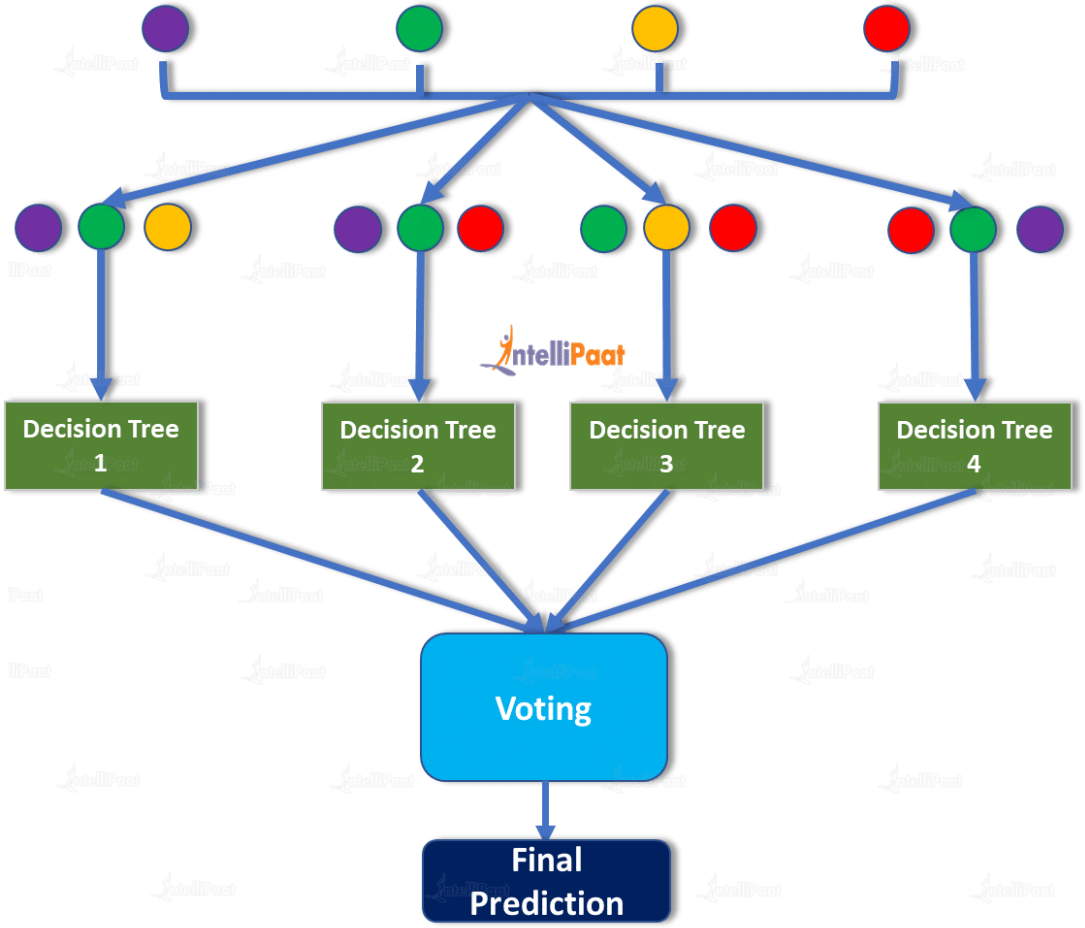
Random Forest

- A machine learning algorithm used for classification and regression
- Based on combining multiple decision trees
- Known as an ensemble method (uses multiple models to improve performance)

How Does It Work?

- Draw multiple random samples from data
- Fit tree to each sample using a random set of predictors
- Combine the classifications/predictions from all the resampled trees (the “forest”).

Random Forest



Random Forest Example